

Statistics for Data Science-1

Week-9 Graded Assignment

1. A discrete random variable X can take the values $1, 2, 3, \dots, n$. For these values the cumulative distribution function is defined by:

$$F(x) = P(X \leq x) = \frac{x^2 + k}{m} ; x = 1, 2, 3, \dots, n$$

Find the value of k .

Answer: $k = m - n^2$

Solution:

$$F(n) = P(X \leq n) = 1$$

$$\Rightarrow \frac{n^2 + k}{m} = 1$$

Hence, $k = m - n^2$

Suppose, we substitute values of n and m as 3 and 40 respectively, then

$$\frac{3^2 + k}{40} = 1$$

$$k = 31$$

2. An organization in Texas organizes a lucky draw this month. n thousand tickets are sold for m \$ each. Each has an equal chance of winning. x tickets will win a \$, y tickets will win b \$ and z tickets will win c \$. Let, the random variable X denote the net gain from purchase of one ticket. What is the probability that X takes a value less than b ? (Enter the answer correct to 4 decimal place)

Answer: $\frac{n \times 1000 - x}{n \times 1000}$

Solution:

X can take values $-m, c - m, b - m$ and $a - m$

$$P(X < b) = P(X = b - m) + P(X = c - m) + P(X = -m)$$

$$P(X < b) = \frac{y}{n \times 1000} + \frac{z}{n \times 1000} + \frac{n \times 1000 - x - y - z}{n \times 1000}$$

$$P(X < b) = \frac{n \times 1000 - x}{n \times 1000}$$

Suppose, we substitute values of n, m, x, a, y, b, z and c as 5, 1, 1, 1000, 2 500, 10 and 100 respectively, then

$$P(X < 500) = P(X = 499) + P(X = 99) + P(X = -1) = \frac{2}{5000} + \frac{10}{5000} + \frac{4987}{5000}$$

$$\text{Therefore, } P(X < 500) = \frac{4999}{5000} = 0.9998$$

3. In a group of n people, x are photographers and $n - x$ are journalists. m people are randomly picked from a group of these n people. Let, Y be a random variable which represents the number of photographers. How many possible values can the random variable Y take?

Answer: $m + 1$

Solution:

Possible values of Y are $0, 1, 2, \dots, m$.

Hence, the number of possible values Y can take is $m + 1$.

Suppose, we substitute values of m , x and n as 8, 240 and 15 respectively, then possible values of Y are $0, 1, 2, \dots, 8$

Hence, the number of possible values Y can take is 9.

4. Which of the following is/are discrete random variables?
- a. Number of tires produced in an automotive tire factory every 30 minutes.
 - b. The number of kernels(pieces) of popcorn in a 1 *kg* container.
 - c. The time between customers entering a checkout lane at a retail store.
 - d. The amount of rain recorded at an airport one day.
 - e. The amount of liquid in a 2 *litres* bottle of soft drink.
 - f. The number of no-shows for every 1000 reservations made with a commercial airline.

Answer: a, b, f

Solution:

The number of tires produced in an automotive tire factory every 30 minutes can have countable possible values, and hence it denotes a discrete random variable.

Hence, option (a) is correct.

The number of kernels of popcorn in a 1 *kg* container also have countable possible values, it cannot take all values between some interval and hence it is a discrete random variable. So option (b) is correct.

The time between customers entering a checkout lane at a retail store can take any values between some interval. Hence, it is a continuous random variable.

So, option (c) is incorrect.

Again, the amount of rain recorded at an airport one day and the amount of liquid in a 2 *litres* bottle of soft drink can take any values between some interval. Hence, they are continuous random variable.

So, option (d) and (e) are incorrect.

The number of no-shows for every 1000 reservations made with a commercial airline can have countable possible values, and hence it denotes a discrete random variable. Hence, option (f) is correct.

5. A biased coin with probability of heads 0.75 is tossed three times. Let X be a random variable that represents the number of head runs, a head run being defined as a consecutive occurrence of at least two heads. Then the probability mass function of X is given by:

a.

$$P(X = x) = \begin{cases} 0.375 & \text{for } x = 0 \\ 0.625 & \text{for } x = 1 \end{cases}$$

b.

$$P(X = x) = \begin{cases} 0.297 & \text{for } x = 0 \\ 0.703 & \text{for } x = 1 \end{cases}$$

c.

$$P(X = x) = \begin{cases} 0.016 & \text{for } x = 0 \\ 0.140 & \text{for } x = 1 \\ 0.422 & \text{for } x = 2 \\ 0.422 & \text{for } x = 3 \end{cases}$$

d.

$$P(X = x) = \begin{cases} 0.016 & \text{for } x = 0 \\ 0.844 & \text{for } x = 1 \\ 0.140 & \text{for } x = 2 \end{cases}$$

Answer: b

Solution:

Possible outcomes	X	$P(X = x)$
HHH	1	0.422
HHT	1	0.141
HTH	0	0.141
HTT	0	0.047
THH	1	0.141
THT	0	0.047
TTH	0	0.047
TTT	0	0.016

Table 9.1

Hence, the probability mass function of X is given by:

$$P(X = x) = \begin{cases} 0.297 & \text{for } x = 0 \\ 0.703 & \text{for } x = 1 \end{cases}$$

6. Nina has n music sessions in a week. She attends the sessions n days a week $x\%$ of the time, $n - 1$ days $y\%$ of the time, one day $z\%$ of the time, and no days $p\%$ of the time. Let, X be a discrete random variable representing the number of sessions she attends in a week. Suppose one week is randomly selected, what is the probability that the random variable X takes the value at most $n - 1$? (Enter the answer correct to 2 decimal places)

Answer: $1 - \frac{x}{100}$

Solution:

The pmf of random variable X is given by:

$$P(X = k) = \begin{cases} \frac{x}{100} & \text{for } k = n \\ \frac{y}{100} & \text{for } k = n - 1 \\ \frac{z}{100} & \text{for } k = 1 \\ \frac{p}{100} & \text{for } k = 0 \end{cases}$$

$$\begin{aligned} P(X \leq n - 1) &= P(X = 0) + P(X = 1) + P(X = n - 1) \\ &= \frac{p}{100} + \frac{z}{100} + \frac{y}{100} \\ &= \frac{p + y + z}{100} \\ &= 1 - \frac{x}{100} \end{aligned}$$

Suppose, we substitute values of n , x , y , z and p as 5, 50, 20, 10 and 20 respectively, then

The pmf of random variable X is given by:

$$P(X = k) = \begin{cases} 0.5 & \text{for } k=5 \\ 0.2 & \text{for } k=4 \\ 0.1 & \text{for } k=1 \\ 0.2 & \text{for } k=0 \end{cases}$$

$$\begin{aligned}
 P(X \leq 4) &= P(X = 0) + P(X = 1) + P(X = 4) \\
 &= 0.2 + 0.1 + 0.2 \\
 &= 0.5
 \end{aligned}$$

7. Find the value of k for which $k \left(\frac{m}{n}\right)^x$ ($x = 0, 1, 2, \dots$) is a pmf. (Enter the answer correct up to 2 decimal places)

Answer: $\frac{n-m}{n}$

Solution:

For pmf: $k \left[\left(\frac{m}{n}\right)^0 + \left(\frac{m}{n}\right)^1 + \left(\frac{m}{n}\right)^2 + \dots \right] = 1$

$$\Rightarrow k \cdot \frac{1}{1 - \frac{m}{n}} = 1$$

$$\Rightarrow k \cdot \frac{n}{n-m} = 1$$

Therefore, $k = \frac{n-m}{n}$.

For example:

Take $m = 3$ and $n = 8$. For pmf: $k \left[\left(\frac{3}{8}\right)^0 + \left(\frac{3}{8}\right)^1 + \left(\frac{3}{8}\right)^2 + \dots \right] = 1$

$$\Rightarrow k \cdot \frac{1}{1 - \frac{3}{8}} = 1$$

$$\Rightarrow k \cdot \frac{8}{5} = 1$$

Therefore, $k = \frac{5}{8}$.

8. Using the information in the previous question, calculate $P(X = 2)$. (Enter the answer correct up to 2 decimal places)

Answer: $\frac{(n-m)}{n} \cdot \left(\frac{m}{n}\right)^2$

Solution:

$$P(X = 2) = \frac{(n-m)}{n} \cdot \left(\frac{m}{n}\right)^2.$$

For example:

Take $m = 3$ and $n = 8$. For pmf: $k \left[\left(\frac{3}{8}\right)^0 + \left(\frac{3}{8}\right)^1 + \left(\frac{3}{8}\right)^2 + \dots \right] = 1$

$$\Rightarrow k \cdot \frac{1}{1 - \frac{3}{8}} = 1$$

$$\Rightarrow k \cdot \frac{8}{5} = 1$$

Therefore, $k = \frac{5}{8}$.

$$\text{And, } P(X = 2) = \frac{5}{8} \cdot \left(\frac{3}{8}\right)^2 = 0.09.$$

9. From a box A containing 3 white and 6 black balls, 5 balls are transferred into an empty box B . Let X be a random variable that represents the number of white balls which are transferred from A to B . What value of random variable will have the least probability?

Answer: 0

Solution:

Let us define the following cases:

Transfer of 0 white and 5 black balls.

Transfer of 1 white and 4 black balls.

Transfer of 2 white and 3 black balls.

Transfer of 3 white and 2 black balls.

Probabilities for all cases:

$$(i) P(X = 0) = \frac{{}^6C_5}{{}^9C_5} = 0.048$$

$$(ii) P(X = 1) = \frac{{}^3C_1 {}^6C_4}{{}^9C_5} = 0.357$$

$$(iii) P(X = 2) = \frac{{}^3C_2 {}^6C_3}{{}^9C_5} = 0.476$$

$$(iv) P(X = 3) = \frac{{}^3C_3 {}^6C_2}{{}^9C_5} = 0.119$$

Thus, $X = 0$ has the least probability.

10. The probability mass function of a random variable X is given by:

$$P(X = x) = \begin{cases} 3k^2 - 3k & \text{for } x = 0 \\ 2k^2 - 1 & \text{for } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine the value of k given $k > 0$.

Answer: 1

Solution:

From properties of pmf,

$$p(0) + p(1) = 1$$

$$3k^2 - 3k + 2k^2 - 1 = 1$$

$$5k^2 - 3k - 2 = 0$$

$$5k^2 - 5k + 2k - 2 = 0$$

$$5k(k - 1) + 2(k - 1) = 0$$

$$(5k + 2)(k - 1) = 0$$

$$k = \frac{-2}{5} \text{ or } k = 1$$

As $k > 0$, therefore $k = 1$.